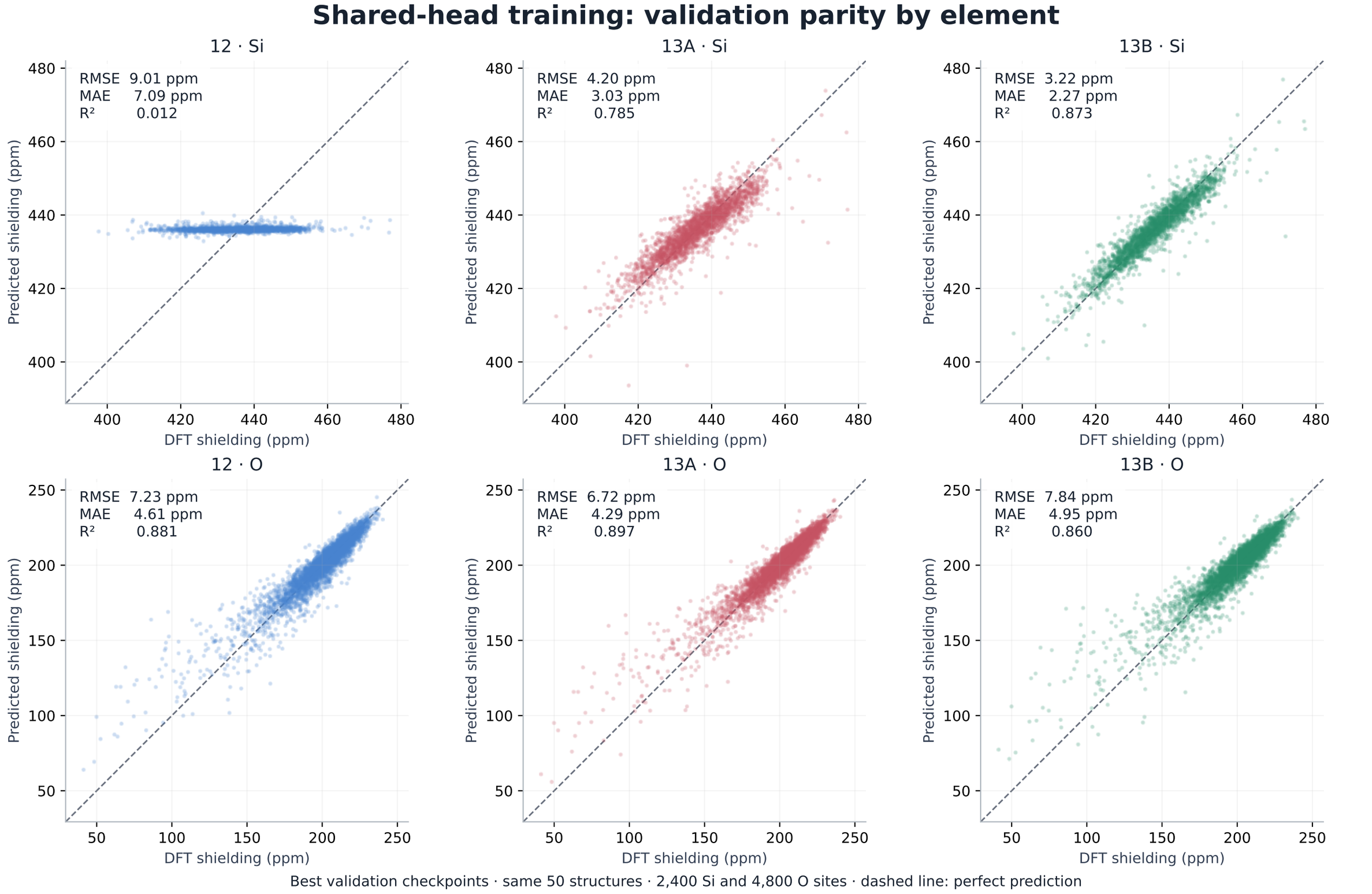

Shared-head training: validation parity by element
12 · Si
RMSE 9.01 ppm
MAE 7.09 ppm
R² 0.012
13A · Si
RMSE 4.20 ppm
MAE 3.03 ppm
R² 0.785
13B · Si
RMSE 3.22 ppm
MAE 2.27 ppm
R² 0.873
12 · O
RMSE 7.23 ppm
MAE 4.61 ppm
R² 0.881
13A · O
RMSE 6.72 ppm
MAE 4.29 ppm
R² 0.897
13B · O
RMSE 7.84 ppm
MAE 4.95 ppm
R² 0.860
Predicted shielding (ppm)
DFT shielding (ppm)
Best validation checkpoints · same 50 structures · 2,400 Si and 4,800 O sites · dashed line: perfect prediction